

$$c_2 < 4$$

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### Abstract

Let  $c_2$  be the least weak-type  $(1, 1)$  constant for the centered Hardy–Littlewood maximal operator over axis-parallel squares in  $\mathbb{R}^2$ . A classical covering argument gives  $c_2 \leq 4$ . We describe an AI-assisted construction, prompted by the human insight to adapt the partial-balayage method of Spector and Stockdale [1], that proves  $c_2 < 4$ .

## 1 Introduction

For  $f \in L^1(\mathbb{R}^2)$  define the centered square maximal operator

$$M_{\square}f(x) = \sup_{r>0} \frac{1}{4r^2} \int_{x+[-r,r]^2} |f(y)| dy, \quad c_2 = \sup_{f \neq 0, \lambda > 0} \frac{\lambda |\{M_{\square}f > \lambda\}|}{\|f\|_1}.$$

The classical covering argument gives  $c_2 \leq 4$ ; see Tao [4]. This article aims at explaining the main idea that leads to  $c_2 < 4$ . The motivating human insight was to adapt the partial-balayage method used by Spector and Stockdale in a different maximal-function problem [1].

Let  $D = \{(u, v) : |u| + |v| \leq 1\}$  be the unit diamond, of area 2, and define

$$M_D g(z) = \sup_{s>0} \frac{1}{2s^2} \int_{z+sD} |g(w)| dw.$$

The linear map  $(u, v) \mapsto (u + v, u - v)$  carries  $D$  onto  $[-1, 1]^2$  and has determinant of absolute value 2. The corresponding change of variables preserves the optimal weak-type  $(1, 1)$  constant, so it suffices to bound  $M_D$ . We do this by dominating  $M_D$  with a suitable convolution maximal operator and proving a weak-type estimate for that operator.

## 2 Kernel comparison

For  $0 < \alpha < 2$ , let  $A_{\alpha} = -|D_u|^{\alpha} - |D_v|^{\alpha}$ , with Fourier symbol  $-(|\xi_1|^{\alpha} + |\xi_2|^{\alpha})$ . For an integrable kernel  $K$ , the distribution  $A_{\alpha}K$  is defined by duality:

$$\langle A_{\alpha}K, \varphi \rangle := \langle K, A_{\alpha}\varphi \rangle = \int_{\mathbb{R}^2} K(x)(A_{\alpha}\varphi)(x) dx.$$

Thus  $A_{\alpha}K \geq 0$  on  $\mathbb{R}^2 \setminus \{0\}$  means this pairing is nonnegative for every nonnegative smooth test function supported away from the origin.

**Theorem 1.** *Let  $K \in L^1(\mathbb{R}^2)$  be even. Suppose  $K \geq \mathbf{1}_D$  a.e. and  $A_{\alpha}K \geq 0$  on  $\mathbb{R}^2 \setminus \{0\}$ . Then  $c_2 \leq \frac{1}{2} \int_{\mathbb{R}^2} K$ .*

*Proof.* The proof has three steps, and the proof of each step is shown in the lemmas below.

**Step 1.** Show that  $M_D g$  is dominated by  $\frac{1}{2} \sup_{s>0} K_s * |g|$ , where  $K_s(x) = s^{-2} K(x/s)$ .

**Step 2.** For a smooth source  $f \geq 0$  and cap  $\kappa$ , we construct a potential  $u$  and a capped density  $\sigma$  with  $A_\alpha u = \sigma - f$ . This is the partial balayage approach used in [1].

**Step 3.** Use the result in Step 2 to establish a weak-type estimate for the operator  $\frac{1}{2} \sup_{s>0} K_s * (\cdot)$ .  $\square$

**Lemma 1.** If  $K \geq \mathbf{1}_D$  a.e., then  $M_D g \leq \frac{1}{2} \sup_{s>0} K_s * |g|$ .

*Proof.* Since  $K \geq \mathbf{1}_D$ , we have  $K_s \geq s^{-2} \mathbf{1}_{sD}$ . For every  $z \in \mathbb{R}^2$  and  $s > 0$ , the substitution  $y = z - w$  and the symmetry of  $D$  give

$$\begin{aligned} \frac{1}{2s^2} \int_{z+sD} |g(y)| dy &= \frac{1}{2s^2} \int_{sD} |g(z-w)| dw \\ &\leq \frac{1}{2} \int_{\mathbb{R}^2} K_s(w) |g(z-w)| dw \\ &= \frac{1}{2} (K_s * |g|)(z). \end{aligned}$$

Taking the supremum over  $s > 0$  proves the claim.  $\square$

**Lemma 2** (Partial-balayage). Let  $f$  be a nonnegative function in  $C_c^\infty(\mathbb{R}^2)$  and  $\kappa$  a positive constant. There exist nonnegative functions  $u$  and  $\sigma$  such that, with  $\Omega = \{u > 0\}$ ,

$$A_\alpha u = \sigma - f \text{ in } \mathcal{D}'(\mathbb{R}^2), \quad \sigma \leq \kappa \text{ a.e.}, \quad \sigma = \kappa \text{ a.e. on } \Omega, \quad \int_{\mathbb{R}^2} \sigma = \int_{\mathbb{R}^2} f, \quad \kappa |\Omega| \leq \|f\|_1.$$

*Proof.* For a real-valued  $v \in L^1(\mathbb{R}^2)$ , define the Dirichlet energy, possibly equal to  $+\infty$ , by

$$\mathcal{E}(v, v) = \int_{\mathbb{R}^2} (|\xi_1|^\alpha + |\xi_2|^\alpha) |\widehat{v}(\xi)|^2 d\xi, \quad \mathcal{C} = \{v \in L^1(\mathbb{R}^2) : v \geq 0, \mathcal{E}(v, v) < \infty\}.$$

$\mathcal{E}$  has the following bilinear representation for  $v, w \in L^1(\mathbb{R}^2)$  of finite Dirichlet energy:

$$\mathcal{E}(v, w) = \frac{c_\alpha}{2} \sum_{j=1}^2 \int_{\mathbb{R}^2} \int_{\mathbb{R}} \frac{[v(x + se_j) - v(x)][w(x + se_j) - w(x)]}{|s|^{1+\alpha}} ds dx.$$

Here  $e_j$  is the  $j$ th coordinate vector, and  $c_\alpha > 0$  is chosen so that  $c_\alpha \int_{\mathbb{R}} (1 - \cos(s\tau)) |s|^{-1-\alpha} ds = |\tau|^\alpha$ . To justify the formula, note first that  $v \in L^1$  implies  $\widehat{v}$  is bounded near the origin; finite energy controls  $\widehat{v}$  away from the origin. Hence  $v \in L^2$ , and likewise  $w \in L^2$ . Plancherel and Tonelli, applied first with  $w = v$ , give the formula for  $\mathcal{E}(v, v)$ ; polarization gives the displayed bilinear identity.

We will prove the desired result by finding a minimizer in  $\mathcal{C}$  of

$$J(v) = \frac{1}{2} \mathcal{E}(v, v) + \kappa \int_{\mathbb{R}^2} v - \int_{\mathbb{R}^2} f v.$$

**Step 1: coercivity and existence of a minimizer.** For every admissible  $v$ , the fractional Sobolev inequality, followed by Hölder's and Young's inequalities, gives

$$\|v\|_p \leq C_\alpha \mathcal{E}(v, v)^{1/2}, \quad \left| \int f v \right| \leq C_\alpha \|f\|_{p'} \mathcal{E}(v, v)^{1/2} \leq \frac{1}{4} \mathcal{E}(v, v) + C_\alpha^2 \|f\|_{p'}^2,$$

where  $p = 4/(2 - \alpha) > 2$  and  $p' = 4/(2 + \alpha)$ ; see [2]. Consequently,

$$J(v) \geq \frac{1}{4}\mathcal{E}(v, v) + \kappa\|v\|_1 - C_\alpha^2\|f\|_{p'}^2 \geq -C_\alpha^2\|f\|_{p'}^2.$$

Thus  $\inf J > -\infty$ , and a minimizing sequence  $(u_n)$  is bounded in energy and in  $L^1$ . Interpolation also bounds its  $L^2$  norm. Hence it is bounded in  $H^{\alpha/2}(\mathbb{R}^2)$ . Weak compactness and local Rellich compactness yield a subsequence converging weakly in  $H^{\alpha/2}(\mathbb{R}^2)$ , strongly in  $L_{\text{loc}}^2$ , and almost everywhere to  $u$ . Since  $u_n \geq 0$ , we have  $u \geq 0$  almost everywhere. Fatou's lemma gives  $\int u \leq \liminf_n \int u_n$  and thus  $u \in L^1$ , and weak lower semicontinuity of the squared  $L^2$  norm of  $(|\xi_1|^\alpha + |\xi_2|^\alpha)^{1/2}\hat{u}_n$  gives  $\mathcal{E}(u, u) \leq \liminf_n \mathcal{E}(u_n, u_n)$ . Finally,  $\int f u_n \rightarrow \int f u$  by local  $L^2$  convergence and compact support of  $f$ . Thus  $J(u) \leq \liminf_n J(u_n) = \inf J$ , proving that  $u$  is a minimizer.

**Step 2: derivation of the first variational inequality.** Fix  $v \in \mathcal{C}$ , put  $w = v - u$ , and set  $u_t = u + tw = (1 - t)u + tv$  for  $0 \leq t \leq 1$ . Convexity of  $\mathcal{C}$  makes  $u_t$  admissible. Minimality gives  $J(u_t) - J(u) \geq 0$ . Therefore, for  $t > 0$ ,

$$0 \leq \frac{J(u_t) - J(u)}{t} = \mathcal{E}(u, v - u) + \int_{\mathbb{R}^2} (\kappa - f)(v - u) + \frac{t}{2}\mathcal{E}(v - u, v - u).$$

Letting  $t \downarrow 0$  proves the first variational inequality

$$\mathcal{E}(u, v - u) + \int_{\mathbb{R}^2} (\kappa - f)(v - u) \geq 0. \quad (1)$$

**Step 3: the Lewy–Stampacchia estimate.** Taking  $v = u + \varphi$  in (1), for  $\varphi \geq 0$ , gives

$$\langle A_\alpha u, \varphi \rangle = -\mathcal{E}(u, \varphi) \leq \int (\kappa - f)\varphi.$$

Thus,  $A_\alpha u \leq \kappa - f$  as a distribution. We now aim to obtain a lower bound. Fix  $t > 0$  and put  $w = (t\varphi - u)_+$ . Positive-part truncation preserves integrability and finite energy, so  $v = (u - t\varphi)_+ = u - t\varphi + w$  is admissible. Equation (1) gives

$$\mathcal{E}(u, t\varphi - w) + \int (\kappa - f)(t\varphi - w) \leq 0.$$

Write  $h = t\varphi - u$ , so  $w = h_+$ . Applying the elementary inequality  $(a - b)(a_+ - b_+) \geq (a_+ - b_+)^2$  to the bilinear representation of  $\mathcal{E}$  gives

$$\begin{aligned} \mathcal{E}(t\varphi - u, w) &= \frac{c_\alpha}{2} \sum_{j=1}^2 \int_{\mathbb{R}^2} \int_{\mathbb{R}} \frac{[h(x + se_j) - h(x)][h_+(x + se_j) - h_+(x)]}{|s|^{1+\alpha}} ds dx \\ &\geq \frac{c_\alpha}{2} \sum_{j=1}^2 \int_{\mathbb{R}^2} \int_{\mathbb{R}} \frac{|h_+(x + se_j) - h_+(x)|^2}{|s|^{1+\alpha}} ds dx \\ &= \mathcal{E}(w, w). \end{aligned}$$

Using this truncation inequality and Cauchy–Schwarz, we obtain

$$\begin{aligned}
\mathcal{E}(u, t\varphi - w) &= t\mathcal{E}(u, \varphi) - \mathcal{E}(u, w) \\
&= t\mathcal{E}(u, \varphi) - t\mathcal{E}(\varphi, w) + \mathcal{E}(t\varphi - u, w) \\
&\geq t\mathcal{E}(u, \varphi) - t\mathcal{E}(\varphi, w) + \mathcal{E}(w, w) \\
&\geq t\mathcal{E}(u, \varphi) - t\mathcal{E}(\varphi, \varphi)^{1/2}\mathcal{E}(w, w)^{1/2} + \mathcal{E}(w, w) \\
&= t\mathcal{E}(u, \varphi) - \frac{t^2}{4}\mathcal{E}(\varphi, \varphi) + \left(\mathcal{E}(w, w)^{1/2} - \frac{t}{2}\mathcal{E}(\varphi, \varphi)^{1/2}\right)^2 \\
&\geq t\mathcal{E}(u, \varphi) - \frac{t^2}{4}\mathcal{E}(\varphi, \varphi).
\end{aligned}$$

Since  $0 \leq t\varphi - w \leq t\varphi$  and  $f \geq 0$ , we also have

$$\begin{aligned}
\int (\kappa - f)(t\varphi - w) &= \kappa \int (t\varphi - w) - \int f(t\varphi - w) \\
&\geq - \int f(t\varphi - w) \\
&\geq -t \int f\varphi.
\end{aligned}$$

Substituting these bounds into the variational inequality above gives

$$\begin{aligned}
0 &\geq \mathcal{E}(u, t\varphi - w) + \int (\kappa - f)(t\varphi - w) \\
&\geq t\mathcal{E}(u, \varphi) - \frac{t^2}{4}\mathcal{E}(\varphi, \varphi) - t \int f\varphi.
\end{aligned}$$

Dividing by  $t > 0$  yields  $\mathcal{E}(u, \varphi) \leq \int f\varphi + \frac{t}{4}\mathcal{E}(\varphi, \varphi)$ . Letting  $t \downarrow 0$  therefore proves  $\mathcal{E}(u, \varphi) \leq \int f\varphi$ . Together, the two bounds prove

$$-f \leq A_\alpha u \leq \kappa - f \quad \text{in } \mathcal{D}'(\mathbb{R}^2). \quad (2)$$

Define  $\sigma = f + A_\alpha u$ , which is a distribution. By (2), it is positive and bounded above by the distribution induced by the constant function  $\kappa$ . A positive distribution is a nonnegative Radon measure; the bound  $\sigma \leq \kappa dx$  then makes it absolutely continuous. Its Radon–Nikodym density, also denoted by  $\sigma$ , satisfies  $0 \leq \sigma \leq \kappa$  almost everywhere. Thus  $A_\alpha u = \sigma - f$  holds distributionally with a bounded function on the right.

**Step 4: equality on the positivity set.** Taking  $v = 0$  and  $v = 2u$  in (1) gives opposite inequalities, hence

$$\mathcal{E}(u, u) + \kappa \int u - \int fu = 0.$$

The identity  $A_\alpha u = \sigma - f$  also gives  $\mathcal{E}(u, u) = \int u(f - \sigma)$ . To justify this test, approximate  $u$  by smooth compactly supported functions in both  $L^1$  and the energy norm, using cutoffs and convolution; the pairings converge because  $f - \sigma$  is bounded. Subtracting the identities yields

$$\int u(\kappa - \sigma) = 0.$$

The integrand is nonnegative, so  $\sigma = \kappa$  almost everywhere on  $\Omega = \{u > 0\}$ .

**Step 5: conservation of mass.** Choose a smooth radial cutoff  $0 \leq \chi \leq 1$ , equal to one on the unit ball, compactly supported, and nonincreasing in the radius. Then  $\chi_R(x) = \chi(x/R) \uparrow 1$ , and homogeneity gives  $\|A_\alpha \chi_R\|_\infty = R^{-\alpha} \|A_\alpha \chi\|_\infty$ . Testing  $A_\alpha u = \sigma - f$  against  $\chi_R$  gives

$$\int \sigma \chi_R = \int f \chi_R + \int u A_\alpha \chi_R, \quad \left| \int u A_\alpha \chi_R \right| \leq R^{-\alpha} \|u\|_1 \|A_\alpha \chi\|_\infty \longrightarrow 0.$$

Monotone convergence, using  $\sigma \geq 0$ , proves both its integrability and  $\int \sigma = \int f$ . Finally, Step 4 and conservation of mass yield the area estimate:

$$\kappa |\Omega| = \int_\Omega \sigma \leq \int \sigma = \|f\|_1.$$

□

**Lemma 3** (Kernel transfer from the obstacle). *Under the hypotheses of Theorem 1, for every nonnegative  $f \in C_c^\infty(\mathbb{R}^2)$  and  $\lambda > 0$ ,*

$$\left| \left\{ x : \sup_{s>0} (K_s * f)(x) > \lambda \right\} \right| \leq \frac{\int_{\mathbb{R}^2} K}{\lambda} \|f\|_1.$$

*Proof.* Choose  $\kappa = \lambda / \int_{\mathbb{R}^2} K$ . The denominator is positive since  $K \geq \mathbf{1}_D$ . Lemma 2 supplies  $u, \sigma$  and  $\Omega = \{u > 0\}$  with

$$A_\alpha u = \sigma - f, \quad 0 \leq \sigma \leq \kappa, \quad |\Omega| \leq \frac{\|f\|_1}{\kappa}.$$

We will prove that  $K_s * f \leq \lambda$  almost everywhere outside  $\Omega$  for every scale, and then bound the superlevel set by  $|\Omega|$ .

**Step 1: the positive minimum principle for  $B$ .** Define the operator  $B$  on smooth compactly supported functions by  $B\varphi = K * (A_\alpha \varphi)$ . Its distributional convolution kernel is  $A_\alpha K$ . We first prove that  $B$  satisfies the *positive-minimum principle*. That is, we assume  $\varphi \in C_c^\infty(\mathbb{R}^2)$ ,  $\varphi \geq 0$ , and  $\varphi(x_0) = 0$ , and we want to prove that  $B\varphi(x_0) \geq 0$ . Since convolution and the Fourier multiplier commute with translations, we may assume without loss of generality that  $x_0 = 0$ . Evenness of  $K$  gives  $B\varphi(0) = \langle A_\alpha K, \varphi \rangle$ . The assumed sign of  $A_\alpha K$  applies only to tests supported away from zero. To remove this restriction, choose  $\eta \in C_c^\infty(\mathbb{R}^2)$  with  $0 \leq \eta \leq 1$  and  $\eta = 1$  near zero, and put  $g_\varepsilon(x) = \eta(x/\varepsilon)\varphi(x)$ . Since zero is a minimum of  $\varphi$ , we have  $\nabla \varphi(0) = 0$ . Taylor's theorem gives  $|\varphi(x)| \leq C|x|^2$  and  $|\nabla \varphi(x)| \leq C|x|$  near zero. The product rule therefore yields

$$\|g_\varepsilon\|_\infty \leq C\varepsilon^2, \quad \|D^2 g_\varepsilon\|_\infty \leq C.$$

$A_\alpha g_\varepsilon$  has the following integral representation:

$$A_\alpha g_\varepsilon(x) = \frac{c_\alpha}{2} \sum_{j=1}^2 \int_{\mathbb{R}} \frac{g_\varepsilon(x + te_j) + g_\varepsilon(x - te_j) - 2g_\varepsilon(x)}{|t|^{1+\alpha}} dt,$$

where  $c_\alpha$  is a positive constant. The numerator is bounded by  $Ct^2$  for  $|t| \leq \varepsilon$ , using the second-derivative bound, and by  $C\varepsilon^2$  for  $|t| > \varepsilon$ , using the bound on  $g_\varepsilon$ . Hence

$$\|A_\alpha g_\varepsilon\|_\infty \leq C \left( \int_0^\varepsilon t^{1-\alpha} dt + \varepsilon^2 \int_\varepsilon^\infty t^{-1-\alpha} dt \right) \leq C_\alpha \varepsilon^{2-\alpha} \longrightarrow 0.$$

Both integrals are finite because  $0 < \alpha < 2$ . Since  $K \in L^1$ ,

$$|\langle A_\alpha K, g_\varepsilon \rangle| = \left| \int K A_\alpha g_\varepsilon \right| \leq \left( \int_{\mathbb{R}^2} K \right) \|A_\alpha g_\varepsilon\|_\infty \longrightarrow 0.$$

The function  $\varphi - g_\varepsilon = (1 - \eta(\cdot/\varepsilon))\varphi$  is nonnegative and vanishes near zero. Therefore  $\langle A_\alpha K, \varphi - g_\varepsilon \rangle \geq 0$ , and passage to the limit proves  $B\varphi(0) \geq 0$ .

**Step 2: application of the Courrège representation theorem.** We now show that  $B$  also satisfies the *positive-maximum principle*. This means that if  $\varphi \in C_c^\infty(\mathbb{R}^2)$  attains a nonnegative global maximum at  $x$ , then  $B\varphi(x) \leq 0$ . Let this maximum be  $m$ . Choose a smooth cutoff  $0 \leq \chi_R \leq 1$  that is equal to one on a ball containing both  $\text{supp } \varphi$  and  $x$ . Then

$$m\chi_R - \varphi \geq 0, \quad (m\chi_R - \varphi)(x) = 0.$$

The positive-minimum principle proved in Step 1, applied at the point  $x$ , gives  $B\varphi(x) \leq mB\chi_R(x)$ . Writing  $\chi_R(y) = \chi(y/R)$ , homogeneity of  $A_\alpha$  gives

$$\|B\chi_R\|_\infty = \|K * A_\alpha \chi_R\|_\infty \leq \|K\|_1 \|A_\alpha \chi_R\|_\infty = \left( \int_{\mathbb{R}^2} K \right) R^{-\alpha} \|A_\alpha \chi\|_\infty \longrightarrow 0.$$

Letting  $R \rightarrow \infty$  proves  $B\varphi(x) \leq 0$ .

Since  $B$  is translation invariant and satisfies the positive-maximum principle, the Courrège representation theorem applies [3]:

$$B\varphi(x) = -c\varphi(x) + b \cdot \nabla \varphi(x) + \text{tr}(QD^2\varphi(x)) + \int_{\mathbb{R}^2 \setminus \{0\}} (\varphi(x+z) - \varphi(x) - \mathbf{1}_{\{|z|<1\}} z \cdot \nabla \varphi(x)) \nu(dz),$$

where  $Q$  is positive semidefinite,  $c \geq 0$ , and  $\nu$  is a nonnegative measure satisfying

$$\int_{\mathbb{R}^2 \setminus \{0\}} \min(1, |z|^2) \nu(dz) < \infty.$$

We now remove the terms in the representation one at a time.

- *Zeroth-order term.* Use the cutoffs  $\chi_R(x) = \chi(x/R)$ . They equal one near the origin, so  $\chi_R(0) = 1$ ,  $\nabla \chi_R(0) = 0$ , and  $D^2 \chi_R(0) = 0$ . Both derivative terms vanish at  $x = 0$ , without using any symmetry of  $\nu$ . The representation there reads

$$B\chi_R(0) = -c + \int_{\mathbb{R}^2 \setminus \{0\}} (\chi_R(z) - 1) \nu(dz).$$

The integral vanishes on  $|z| \leq R$  and its absolute value is at most  $\nu(\{|z| > R\})$ , which tends to zero because  $\nu$  has finite mass outside the unit ball. The estimate  $\|B\chi_R\|_\infty \rightarrow 0$  proved above now gives  $c = 0$ .

- *First-order term and symmetry of the measure.* Let  $(R\varphi)(x) = \varphi(-x)$ . Evenness of  $K$  and of  $A_\alpha$  gives  $RBR = B$ . Reflecting the displayed Courrège formula replaces  $b$  by  $-b$  and  $\nu$  by its reflected measure  $\nu^-$ , defined by  $\nu^-(E) = \nu(-E)$  for every Borel set  $E$ . Since  $B = RBR$ , we may average these two formulas. Their first-order terms  $b \cdot \nabla \varphi$  and  $-b \cdot \nabla \varphi$  cancel. The measure becomes  $(\nu + \nu^-)/2$ ; we continue to denote it by  $\nu$ . Thus  $\nu$  is *symmetric*:  $\nu(E) = \nu(-E)$  for every Borel set  $E$ .

- *Second-order term.* The multiplier of  $-B$  is  $(|\xi_1|^\alpha + |\xi_2|^\alpha)\widehat{K}(\xi)$ , whose absolute value is at most  $(|\xi_1|^\alpha + |\xi_2|^\alpha) \left(\int_{\mathbb{R}^2} K\right)$ . The representation, with  $c = 0$  and symmetric  $\nu$ , gives

$$\xi^T Q \xi + \int_{\mathbb{R}^2 \setminus \{0\}} (1 - \cos(\xi \cdot z)) \nu(dz) = (|\xi_1|^\alpha + |\xi_2|^\alpha) \widehat{K}(\xi).$$

Both terms on the left are nonnegative. If  $Q \neq 0$ , choose a unit vector  $e$  with  $e^T Q e > 0$  and set  $\xi = re$ . We would obtain

$$r^2 e^T Q e \leq \left( \int_{\mathbb{R}^2} K \right) r^\alpha (|e_1|^\alpha + |e_2|^\alpha),$$

which is impossible as  $r \rightarrow \infty$  because  $\alpha < 2$ . Hence  $Q = 0$ .

- *Cancellation of the gradient term inside the integral.* Fix  $x$ . Since  $\nu(E) = \nu(-E)$ , the substitution  $z \mapsto -z$  in the integral gives

$$\begin{aligned} B\varphi(x) &= \int [\varphi(x+z) - \varphi(x) - \mathbf{1}_{\{|z|<1\}} z \cdot \nabla \varphi(x)] \nu(dz) \\ &= \int [\varphi(x-z) - \varphi(x) + \mathbf{1}_{\{|z|<1\}} z \cdot \nabla \varphi(x)] \nu(dz). \end{aligned}$$

Averaging these two equal expressions cancels the gradient terms inside the integrand and yields

$$B\varphi(x) = \frac{1}{2} \int_{\mathbb{R}^2 \setminus \{0\}} (\varphi(x+z) + \varphi(x-z) - 2\varphi(x)) \nu(dz).$$

For  $v, w \in C_c^\infty(\mathbb{R}^2)$ , define  $\mathcal{E}_B(v, w) := - \int_{\mathbb{R}^2} v B w$ . Insert the preceding formula for  $Bw$  and use the change of variables formula. This gives

$$\mathcal{E}_B(v, w) = \frac{1}{2} \int_{\mathbb{R}^2 \setminus \{0\}} \int_{\mathbb{R}^2} [v(x+z) - v(x)][w(x+z) - w(x)] dx \nu(dz).$$

This formula also shows  $\mathcal{E}_B(v, w) = \mathcal{E}_B(w, v)$ .

As in Lemma 2, apply  $(a-b)(a_+ - b_+) \geq (a_+ - b_+)^2$  inside this integral to obtain

$$\mathcal{E}_B(v, v_+) \geq \mathcal{E}_B(v_+, v_+).$$

It remains to check that the potential  $u$  belongs to this energy domain. Plancherel's identity and the Fourier multiplier above give

$$\mathcal{E}_B(v, v) = \int_{\mathbb{R}^2} (|\xi_1|^\alpha + |\xi_2|^\alpha) \widehat{K}(\xi) |\widehat{v}(\xi)|^2 d\xi.$$

The Courrège representation makes this integral nonnegative. Moreover,  $\widehat{K}(\xi) \leq \int_{\mathbb{R}^2} K$ . Therefore

$$0 \leq \mathcal{E}_B(v, v) \leq \left( \int_{\mathbb{R}^2} K \right) \mathcal{E}(v, v).$$

Lemma 2 gives  $\mathcal{E}(u, u) < \infty$ , so  $u$  has finite  $\mathcal{E}_B$  energy. This fact is needed in Step 3: it ensures that the truncation  $(\varepsilon\varphi - u)_+$  also has finite  $\mathcal{E}_B$  energy and that every pairing in the truncation argument is well defined.

**Step 3: bound the convolution outside  $\Omega$ .** Convolving the obstacle equation with  $K$  gives

$$Bu = K * \sigma - K * f$$

in distributions. The right-hand side is bounded because  $K \in L^1$  and  $\sigma, f \in L^\infty$ , so  $Bu$  is represented by a bounded function. We show that this function is nonnegative almost everywhere on  $\Omega^c = \{u = 0\}$ .

Fix a nonnegative smooth compactly supported  $\varphi$  and put  $w = (\varepsilon\varphi - u)_+$ , where  $\varepsilon > 0$ . The truncation inequality gives

$$\mathcal{E}_B(u, w) - \varepsilon \mathcal{E}_B(\varphi, w) = \mathcal{E}_B(u - \varepsilon\varphi, w) \leq -\mathcal{E}_B(w, w) \leq 0.$$

Dividing this truncation inequality by  $\varepsilon$  now gives

$$-\int (Bu) \frac{w}{\varepsilon} = \frac{1}{\varepsilon} \mathcal{E}_B(u, w) \leq \mathcal{E}_B(\varphi, w) = -\int (B\varphi)w \leq \|B\varphi\|_\infty \int w \leq \varepsilon \|B\varphi\|_\infty \int \varphi \longrightarrow 0.$$

Moreover,

$$0 \leq \frac{w}{\varepsilon} \leq \varphi, \quad \frac{w}{\varepsilon} \longrightarrow \varphi \mathbf{1}_{\Omega^c} \quad \text{a.e.}$$

Dominated convergence therefore gives  $\int_{\Omega^c} (Bu)\varphi \geq 0$ . Since this holds for every nonnegative test function,  $Bu \geq 0$  almost everywhere on  $\Omega^c$ . On that set,

$$K * f = K * \sigma - Bu \leq K * \sigma \leq \kappa \int_{\mathbb{R}^2} K = \lambda.$$

**Step 4: take the supremum over scales.** Every  $K_s$  is even and nonnegative, has integral  $\int_{\mathbb{R}^2} K_s$ , and satisfies  $A_\alpha K_s \geq 0$  away from zero by homogeneity. Steps 1–3 therefore apply to  $K_s$  with the same  $u, \sigma, \Omega$ , giving  $K_s * f \leq \lambda$  almost everywhere on  $\Omega^c$  for each fixed  $s > 0$ .

For each positive rational scale  $q$ , the preceding argument gives  $(K_q * f)(x) \leq \lambda$  for almost every  $x \in \Omega^c$ . For each positive rational  $q$ , we denote its corresponding exceptional null set by  $N_q$ . Let  $N := \bigcup_{q \in \mathbb{Q}, q > 0} N_q$ . On  $\Omega^c \setminus N$ , the inequality holds for every rational  $q > 0$ . For each  $x$ ,  $(K_s * f)(x)$  is continuous in  $s > 0$ , which can be proved by dominated convergence.

Fix  $x \in \Omega^c \cap N^c$ . Approximating any real  $s > 0$  by positive rational scales and using continuity gives

$$\sup_{s > 0} (K_s * f)(x) \leq \lambda.$$

Thus the superlevel set of the supremum is contained in  $\Omega \cup N$ . Consequently,

$$\left| \left\{ x : \sup_{s > 0} (K_s * f)(x) > \lambda \right\} \right| \leq |\Omega \cup N| = |\Omega| \leq \frac{\|f\|_1}{\kappa} = \frac{\int_{\mathbb{R}^2} K}{\lambda} \|f\|_1.$$

□

### 3 Optimization formulation and candidate kernels

**Theorem 2.**  $c_2 < 4$ .

By Theorem 1, to prove Theorem 2 it is enough to consider

$$C_\alpha = \inf \left\{ \frac{1}{2} \int_{\mathbb{R}^2} K : K \in L^1(\mathbb{R}^2) \text{ is even, } K \geq \mathbf{1}_D, A_\alpha K \geq 0 \text{ on } \mathbb{R}^2 \setminus \{0\} \right\},$$

and to prove that  $C_\alpha < 4$  for some  $\alpha \in (0, 2)$ . There are two natural approaches. The first is to choose a specific value of  $\alpha$ , construct a corresponding kernel  $K$ , and compute  $\frac{1}{2} \int_{\mathbb{R}^2} K$ . This is the approach taken by the AI agents, which produced the following two concrete examples:

- Let  $\alpha = 1$ ,  $r = |u| + |v|$ ,  $R = 3797/2000$ , and  $\varepsilon = 159411/200000$ . Define

$$K(u, v) = \frac{(R/r - 1)_+}{R - 1} - \varepsilon(1 - r)_+^2, \quad r > 0.$$

The first candidate has been verified in Lean:

$$\frac{1}{2} \int_{\mathbb{R}^2} K = \frac{R^2}{R - 1} - \frac{\varepsilon}{6} < 3.878624.$$

Consequently,  $c_2 < 3.878624$ . The Lean code is available on [GitHub](#), and the certificate is registered on [Palomar](#).

- Let  $\alpha = 6/5$ ,  $R = 7/4$ , and  $r = |u| + |v|$ . Define the cubic cardinal spline

$$\beta(x) = \frac{1}{6} \sum_{q=0}^4 (-1)^q \binom{4}{q} (x + 2 - q)_+^3.$$

The kernel is

$$K(u, v) = a(r^{-6/5} - R^{-6/5})_+ + \sum_{i,j} C_{ij} \beta(16u - i) \beta(16v - j), \quad a = \frac{11248245541992991}{6250000000000000}.$$

The AI-assisted computation gives

$$\frac{1}{2} \int_{\mathbb{R}^2} K = \frac{33744736625978973}{12500000000000000} \left(\frac{7}{4}\right)^{4/5} - \frac{3114374068343977777}{5120000000000000000} < 3.615749.$$

This result has not been verified in Lean because the computation is too large: checking it would require excessive CPU time in the Lean kernel.

The second, more ambitious approach is to determine the exact theoretical infimum  $\inf_{0 < \alpha < 2} C_\alpha$ . The AI-assisted search has not yet made substantial progress on this harder problem.

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